

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 21)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

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0 1 . 2

Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

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0 1 . 3

Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

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10

0 2 . 1

Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

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0 2 . 2

Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

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0 2 . 3

Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

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0 2 . 4

Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

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10

03 . 1

Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

03 . 2

Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

03 . 3

Explain how buying fairtrade food affects producers.

[1 mark]

03 . 4

Explain how nitrate from farmland can threaten human health.

[2 marks]

03 . 5

Give **two** agricultural sources of methane.

[2 marks]

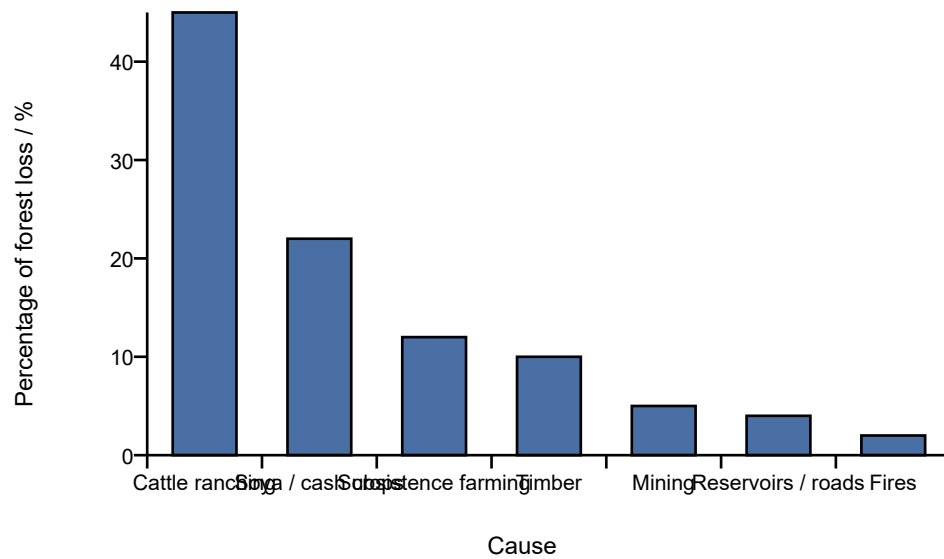
1

2

10

0 4 Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

Figure 1



0 4 . 1 Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

[3 marks]

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04 . 2

Explain why clearance for cattle ranching or cash crops is usually unsustainable.

[2 marks]

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04 . 3

Describe how each of the following threatens rainforest.

[3 marks]

Mineral extraction

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HEP reservoirs

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Climate change

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04 . 4

Describe how a debt-for-nature swap works.

[2 marks]

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10

05

The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

05 . 1

Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

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0 5 . 2

Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

0 5 . 3

Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

10

0	6	.	1
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Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0	6	.	2
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Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

10

07

Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

Category	Population decline over 10 years or 3 generations	Area of habitat	Number of mature adults	Probability of extinction in the wild
Vulnerable	over 40%	under 2000 km ²	under 10 000	over 10% in 100 years
Endangered	over 60%	under 500 km ²	under 2500	over 20% in 20 years or 5 generations
Critically endangered	over 80%	under 10 km ²	under 250	over 50% in 10 years or 3 generations

07**1**

State **three** roles of the IUCN.

[3 marks]

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07**2**

A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km² of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

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07**3**

Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

07 . 4

Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

10

08

A conservation team is planning to protect a population of an endangered antelope in a reserve.

08 . 1

Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size

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Distribution

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Survival rate

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Age structure

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08 . 2 Describe how the population size of the antelope could be estimated.
[2 marks]

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10

09 . 1

A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate

☐

B The readings are accurate but not precise

☐

C The readings are both accurate and precise

☐

D The readings are reliable because the meter is electronic

☐

09 . 2

Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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09 . 3

Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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10 . 1

Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

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[illegible]

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10 . **2** State what is meant by a **taxon**.

[1 mark]

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

OR

1 1 . 2

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS